BABU BANARASI DAS UNIVERSITY

LUCKNOW

Session : 2024-2025

SCHOOL OF COMPUTER APPLICATION

Project
ON

APPLY END TO END CLOUD SECURITY TO APPLICATONS

(BCACSN14201)

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Case Study :- Data Protection and cloud Security.

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1.) Data protection in Healthcare cloud systems :->

A healthcare company stores sensitive patient Data such as medical records, personal details, and billing information in the cloud. Protecting this data is critically due to Privacy laws and cyber threats.

Techniques to protect Data :->

1.) Data Anonymisation

Is the process of removing or modifying personally identifiable information (PII), So that individuals cannot be identified.

How it helps :->

- Removes patient identity.
- Data can still be used for rearch and analysis.
- Reduce risk in case of data breach.

Ex :-

Patient Name -> Removed.

Patient ID :-> Random number

2.) Tokenisation

Tokenisation Replaces sensitive data with a unique

: token (random value) that has no meaningful value outside the system.

How it helps:

original data is stored securely
Tokens are used during processing
Even if stolen, tokens are useless.

Eg:-

Credit card number → Token (eg. X7A92B).

3.) Data Loss Prevention (DLP)

DLP prevents unauthorized access or sharing of sensitive data.

use:-

- Monitors data movement
- Blocks data leaks.

RISKS and Solution

RISK	Description	Solution.
Data Breach	unauthorized access to Patient Data	Encryption + IAM
Insider Threat	Misuse by employee	RBAC + Monitoring
weak Token security	Tokens can be exposed	Secure token Vault.
cloud Misconfiguration	wrong settings expose data	Regular audits.

2.) Network and Infrastructure security.

↳ Cloud organizations face attacks like DDO's, hacking and data interception.

2.1) Technology used

a) Software Defined Networking (SDN)

- Centralized network control
- Dynamic traffic management

b.) VXLAN (Virtual Extensible LAN)

- Network segmentation
- Isolates workloads.

(c.) Intrusion Detection System (IDS)

- Detects suspicious activities
- Alerts administrators

3. Attack mitigation

DDoS Attack

- Load Balancer distributes traffic
- SDN reroutes traffic

Unauthorized access.

IAM + multifactor authentication.

Data interception

User encryption (HTTPS, VPN).

Secure architecture Diagram.

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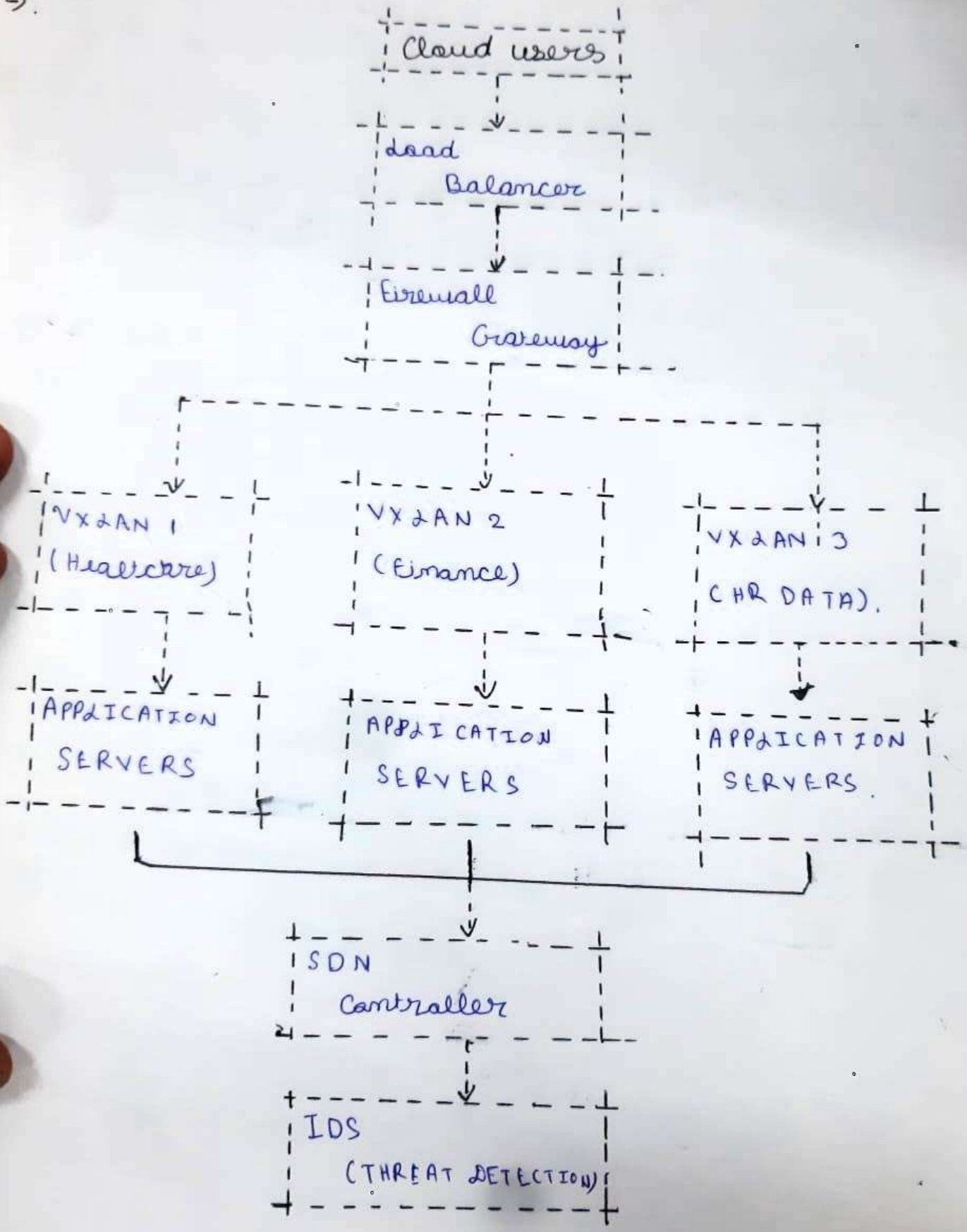

(6) Conclusion

Cloud security is critical in healthcare systems. Data protection techniques like anonymization, tokenization and DLP ensure privacy, while SDN, VXLAN and IDS provide strong network security. A combined approach helps build a secure and reliable cloud environment.